

S-Path-RAG: Semantic-Aware Shortest-Path Retrieval Augmented Generation for Multi-Hop Knowledge Graph Question Answering

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Abstract

We present S-Path-RAG, a semantic-aware shortest-path Retrieval-Augmented Generation framework designed to improve multi-hop question answering over large knowledge graphs. S-Path-RAG departs from one-shot, text-heavy retrieval by enumerating bounded-length, semantically weighted candidate paths using a hybrid weighted k -shortest, beam, and constrained random-walk strategy, learning a differentiable path scorer together with a contrastive path encoder and lightweight verifier, and injecting a compact soft mixture of selected path latents into a language model via cross-attention. The system runs inside an iterative Neural-Socratic Graph Dialogue loop in which concise diagnostic messages produced by the language model are mapped to targeted graph edits or seed expansions, enabling adaptive retrieval when the model expresses uncertainty. This combination yields a retrieval mechanism that is both token-efficient and topology-aware while preserving interpretable path-level traces for diagnostics and intervention. We validate S-Path-RAG on standard multi-hop KGQA benchmarks and through ablations and diagnostic analyses. The results demonstrate consistent improvements in answer accuracy, evidence coverage, and end-to-end efficiency compared to strong graph- and LLM-based baselines. We further analyze trade-offs between semantic weighting, verifier filtering, and iterative updates, and report practical recommendations for deployment under constrained compute and token budgets.

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CCS Concepts

• Information systems → Information extraction; • Computing methodologies → Semantic networks.

Keywords

Semantic-aware retrieval; Retrieval-Augmented Generation; Knowledge Graph Question Answering; Path-based retrieval; Graph neural networks; Cross-attention injection; Iterative retrieval; Soft latent injection

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Resource Availability:

The source code of this paper has been made publicly available at <https://github.com/furonglegend/spathrag>.

1 Introduction

Large language models (LLMs) have advanced the state of the art across many natural language tasks by learning broad semantic priors from large-scale pretraining corpora [21, 47, 52]. Despite their impressive linguistic capabilities, LLMs remain challenged by factual consistency and structured, multi-step reasoning: they are prone to hallucinations and cannot, on their own, reliably traverse large structured knowledge sources to produce fully grounded answers [3, 32]. Knowledge graphs (KGs) store human-curated facts and multi-hop relations in an updatable, structured format and therefore serve as a natural external memory for grounding LLM generation in knowledge-intensive tasks such as question answering (KGQA). Retrieval-augmented generation (RAG) mitigates hallucinations by enriching the model context with retrieved evidence,

but the overall effectiveness of RAG depends fundamentally on the quality of the retriever: for KGQA the retriever must surface multi-hop, relational evidence from very large graphs while avoiding spurious facts that can mislead the generator [29, 32, 58, 59].

Recent work has pursued multiple directions to bridge LLMs and KGs; representative efforts include graph-aware neural retrieval for LLM reasoning, extracting and following relation paths for LLMs, graph instruction tuning, and agentic deep-search workflows for graph retrieval-augmented generation [29, 44, 54, 59]. Existing retrieval strategies for KG-aware RAG can be grouped into two main paradigms. One paradigm depends on the language model itself to propose relation paths or candidate facts: this LLM-driven retrieval can be effective for semantically simple queries but often fails to capture the deep topological evidence required by complex multi-hop questions and can become expensive when many LLM calls are needed [54, 59]. The other paradigm leverages graph-oriented models, principally graph neural networks (GNNs), to reason over subgraphs and surface structurally relevant nodes and paths; while GNNs are strong at topology-aware, multi-hop search, they lack the fine-grained natural language understanding necessary to disambiguate semantically similar candidate paths [8, 13, 29]. Hybrid approaches combine these strengths by using graph algorithms or learned GNNs to produce candidate paths and then presenting those candidates to an LLM, or by injecting graph latents directly into LLMs [3, 6, 44]. Although such hybrids often increase faithfulness and multi-hop coverage, several limitations persist: many systems perform one-shot retrieval and therefore cannot iteratively refine evidence when the LLM signals uncertainty [54, 58]; path enumeration is often agnostic to semantic alignment with the query, producing long textual lists that waste token budget or introduce distracting noise [21, 59]; and the interface between retrieved graph structures and LLMs is frequently ad hoc (raw textual verbalization or coarse latents), which complicates end-to-end optimization and reduces interpretability [9, 28].

To address these limitations, we introduce **S-Path-RAG**, a *semantic-aware shortest-path retrieval augmented generation* framework for multi-hop KGQA. Specifically, S-Path-RAG departs from one-shot, text-heavy retrieval in four complementary ways. First, it explicitly enumerates bounded-length candidate paths using a hybrid retrieval strategy that combines weighted k -shortest path search, beam search, and constrained random walks; this strategy ranks paths by a mixture of structural plausibility, relation priors and learned semantic alignment to the natural-language query. Second, S-Path-RAG trains a differentiable path scorer and a contrastive path encoder together with a lightweight verifier so that paths which are plausible to an LLM but unsupported by the KG receive lower weight. Third, instead of verbalizing long path lists, the system forms a compact soft mixture of selected path latents and injects this latent context into the LLM via cross-attention, allowing the language model to reason over concise structured evidence while keeping token usage low. Fourth, S-Path-RAG operates inside an iterative, LLM-guided retrieval loop (which we call a Neural-Socratic Graph Dialogue) in which diagnostic messages from the LLM are mapped to targeted graph edits or seed expansions and applied as soft-to-discrete graph updates when beneficial.

Our technical contributions are as follows. We propose a semantic-aware shortest-path enumeration method that integrates structural costs, relation priors, and learned semantic matching to rank bounded-length paths for knowledge graph question answering (KGQA) retrieval. To improve answer faithfulness without incurring excessive large language model (LLM) calls, we develop a differentiable path scoring and verification pipeline that jointly trains a contrastive path encoder and a verifier to suppress LLM-plausible false positives. We further introduce soft latent path injection via cross-attention, allowing the LLM to attend to compact path representations instead of long verbalizations. To support this mechanism, we define alignment objectives and conduct causal intervention analyses to verify and enhance the utilization of injected latents. Additionally, we design an iterative diagnostic-to-edit mechanism, termed Neural-Socratic Graph Dialogue, which translates LLM diagnostics into precise graph updates and seed expansions, enabling adaptive retrieval guided by model uncertainty. Finally, we validate S-Path-RAG on standard multi-hop KGQA benchmarks and present comprehensive ablations and diagnostics that isolate the benefits of semantic weighting, verifier filtering, injection alignment, and iterative updates, demonstrating consistent improvements in accuracy, coverage, and efficiency over strong baselines.

2 Related Work

This work lies at the intersection of Retrieval-Augmented Generation (RAG), graph-based reasoning, and large language model (LLM) augmentation. We review the most relevant directions and highlight how S-Path-RAG differs from and complements prior efforts.

2.1 Retrieval-augmented generation for KGQA

RAG techniques ground LLM outputs in external knowledge and have been successfully applied to open-domain QA and specialized domains. Early RAG formulations demonstrated that augmenting generation with retrieved documents improves factuality and breadth; recent work extends this idea by replacing or complementing flat text retrieval with structured graph retrieval for tasks such as customer support and domain-specific QA [23, 27, 33, 53]. Graph-extended RAG variants aim to preserve relational structure during retrieval and thus reduce hallucination, but they often rely on one-shot retrieval that can miss multi-hop evidence needed for complex queries [11, 29]. GraphTrace [31] recently introduced a modular retrieve–rank–generate pipeline that aligns semantic reasoning with structured KG traversal, achieving strong gains on 5–6-hop economic QA.

2.2 Graph neural networks and path reasoning for KGQA

Graph Neural Networks (GNNs) are widely used to encode subgraphs and to surface multi-hop reasoning chains in KGQA. Methods that perform GNN-based subgraph encoding or explicit path extraction have shown strong performance on benchmarks requiring structural reasoning [18, 24, 29]. However, purely GNN-driven pipelines can struggle with natural-language nuance and semantic matching; consequently, hybrid approaches that combine GNNs

for topology-aware retrieval with LLMs for semantic interpretation are becoming popular [32, 50].

2.3 Integrating LLMs with knowledge graphs

Multiple lines of work investigate how to combine the semantic fluency of LLMs with the factual structure of KGs. Approaches range from verbalizing graph facts into prompts to tighter, learned fusion where graph latents are injected into transformer layers [3, 11, 44]. Surveys and roadmaps summarize these trends and emphasize challenges such as modality alignment, scalability, and controllable grounding [32, 61, 64]. Our proposal follows the fusion paradigm but uses compact, *soft* path latents rather than long textual enumerations to reduce context cost and enable differentiable training.

2.4 Path-based retrieval and path scoring

Path-centric methods explicitly retrieve relational chains that connect question entities to candidate answers; these chains serve as interpretable reasoning traces and as focused evidence for LLMs [5, 19, 59]. Prior works use a mix of shortest-path objectives, path ranking, and pruning heuristics to control noise and maintain coverage [5, 29]. S-Path-RAG builds on this lineage by (i) introducing a semantically-weighted bounded-length path search, and (ii) combining differentiable scoring (Gumbel-Softmax relaxations) with a contrastive path encoder and a lightweight verifier to reduce false positives.

2.5 Iterative and agentic retrieval (multi-turn retrieval)

Static, one-shot retrieval limits the ability to refine evidence when the model is uncertain. Iterative retrieval schemes enable the model to issue follow-up retrievals or to request targeted expansions of the knowledge base [54, 58]. Agentic frameworks and RL-based pipelines further optimize multi-turn retrieval policies [19, 26, 51]. Our Neural-Socratic Graph Dialogue (NSGD) is in this family: it lets the LLM emit compact diagnostic messages that are mapped (via a learnable or templated π_{map}) into concrete graph update actions, while preserving a differentiable core for training.

2.6 Summary and Positioning

In contrast to prior one-shot GraphRAG and path-verbalization approaches, **S-Path-RAG** emphasizes semantically weighted path enumeration, differentiable path scoring combined with verifier-filtered latent injection into the language model, and an iterative, LLM-guided graph update loop that bridges soft, differentiable updates with discrete retrieval actions. These design choices are motivated by key failure modes observed in related work, including missed multi-hop evidence, excessive prompt length, and the lack of adaptive retrieval in the presence of model uncertainty.

3 Methodology

We present **S-Path-RAG**, a semantic-aware shortest-path Retrieval-Augmented Generation framework for multi-hop Knowledge Graph Question Answering (KGQA). Departing from the conventional one-shot retrieve–verbalize–LLM pipeline, S-Path-RAG operates

a *dynamic iterative retrieval–reasoning loop*, wherein the language model can emit diagnostic messages that trigger targeted graph updates. To enhance robustness, interpretability, and scalability, our framework introduces several key innovations: an explicit weighted path search strategy; a differentiable path scoring mechanism that incorporates a verifier and injects soft latent representations into the LLM; an explicit diagnostic-to-retrieval mapper π_{map} , which can be either rule-based or learned; and a principled mechanism for transitioning from soft to discrete graph edits, optionally optimized via reinforcement learning. In the following sections, we formalize the notation, define the core components, and present the complete method along with its mathematical formulation and updated algorithm. The necessity of this hybrid weighting is quantified in Section G.

3.1 Notation and problem statement

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a knowledge graph with entities $\mathcal{V}$ and typed edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{R} \times \mathcal{V}$. Given a natural-language question q , the goal is to produce an answer a by reasoning over a compact subgraph of $\mathcal{G}$. The pipeline begins with an entity-linker that returns seed candidates S_0 ; at iteration t the system maintains a working subgraph $G^{(t)}$, a candidate path set $\mathcal{P}^{(t)}$, and graph latents $\mathcal{Z}^{(t)}$.

3.2 Overview: Iterative Retrieval–Reasoning Loop

S-Path-RAG operates for up to T rounds, where each round consists of a sequence of reasoning and retrieval steps. In each round, the current subgraph $G^{(t)}$ is first encoded using a relation-aware GNN to produce node and relation representations $\mathcal{Z}^{(t)}$. A controlled, weighted search is then used to enumerate candidate paths $\mathcal{P}^{(t)}$. These paths are scored using a differentiable mechanism and verified, resulting in a compact selection $\mathcal{P}_{\text{sel}}^{(t)}$. A soft latent mixture $z_{\text{ctx}}^{(t)}$ is injected into the language model via cross-attention, enabling it to reason over the retrieved paths and generate an answer proposal $\hat{a}^{(t)}$ along with a diagnostic message $m^{(t)}$. This message is mapped to graph edits $\Delta^{(t)} = \pi_{\text{map}}(m^{(t)})$, which are applied to produce the updated graph $G^{(t+1)}$ through either discrete modifications or soft masking. The iteration continues until the model’s confidence surpasses a predefined threshold or the maximum number of rounds T is reached.

3.3 Explicit weighted path search

To make path enumeration transparent and controllable we define per-edge semantic-structural weights and a path ranking function.

$$w_e = \alpha \cdot c_{\text{struct}}(e) + \beta \cdot (1 - \text{sim}(\ell_u, \ell_v)) + \gamma \cdot \pi_{\text{rel}}(r), \quad (1)$$

where w_e is the weight for edge $e = (u, r, v)$, $c_{\text{struct}}(e) \geq 0$ is a structural cost (e.g., inverse prior or length penalty), ℓ_u, ℓ_v are node embeddings, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, $\pi_{\text{rel}}(r)$ is an empirical or learnable relation prior, and $\alpha, \beta, \gamma \geq 0$ balance terms.

$$\text{score}(p; q) = - \sum_{e \in p} w_e + \lambda_{\text{sem}} \cdot \text{sem}(p, q), \quad (2)$$

where $\text{score}(p; q)$ ranks path p for query q , the first term is negative path cost (distance), $\text{sem}(p, q)$ is a learned semantic matching

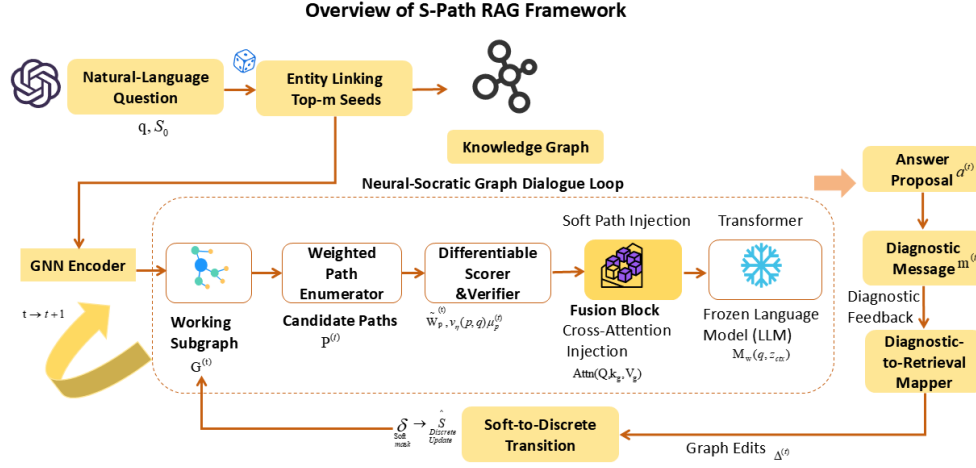


Figure 1: Overview of S-Path-RAG. Given a natural language question, the system performs entity linking to obtain top- m seed entities and enters an iterative Neural-Socratic Graph Dialogue loop. In each round, a GNN encodes the current subgraph, a weighted path enumerator proposes candidate paths, and a scorer with a verifier ranks their relevance. A soft mixture of selected path embeddings is injected into a frozen language model via cross-attention, and the resulting answer together with a diagnostic signal is mapped to graph edits that update the subgraph. The loop terminates when the model reaches sufficient confidence or the maximum number of rounds is reached.

(e.g., cosine between path pooling and query embedding), and λ_{sem} weights semantic relevance.

Candidate generation. We construct $\mathcal{P}^{(t)}$ by a hybrid of (a) k -shortest weighted paths (Yen/Dijkstra), (b) beam-expansion with semantic pruning (beam size B), and (c) top- R random-walk proposals with restart. Practical caps: max length L (typically 3–4), max candidate K (e.g., 50–200). Ablations in experiments should compare weighted k -shortest vs beam vs random-walks.

3.4 Differentiable path scoring and verifier

For each candidate $p \in \mathcal{P}^{(t)}$ compute a learnable score and a soft selection weight:

$$u_p^{(t)} = s_\theta(p, q; \mathcal{Z}^{(t)}), \quad (3)$$

$$\tilde{w}_p^{(t)} = \frac{\exp((u_p^{(t)} + g_p)/\tau)}{\sum_{p' \in \mathcal{P}^{(t)}} \exp((u_{p'}^{(t)} + g_{p'})/\tau)}, \quad (4)$$

where s_θ is a scorer (MLP/Transformer) parameterized by θ using path and graph latents, $g_p \sim \text{Gumbel}(0, 1)$ is Gumbel noise, $\tau > 0$ is temperature, and $\tilde{w}_p^{(t)}$ is the normalized soft weight. Five independent runs yield a coefficient of variation (CV) below 0.5%, confirming the stability of the stochastic scorer. A verifier $v_\eta(p, q) \in [0, 1]$ predicts path utility and is trained with BCE; a contrastive encoder $f_\psi(p, q)$ is trained with InfoNCE to improve semantic ranking. Negatives must include LLM-hard negatives (paths LLM finds plausible).

3.5 Soft path injection and cross-modal fusion

We form a compact injected latent as a weighted mixture:

$$z_{\text{ctx}}^{(t)} = \sum_{p \in \mathcal{P}_{\text{sel}}^{(t)}} \alpha_p^{(t)} \cdot \text{Enc}_{\text{path}}(p), \quad (5)$$

where $\alpha_p^{(t)} \propto \tilde{w}_p^{(t)} \cdot v_\eta(p, q)$, $\text{Enc}_{\text{path}}(p)$ is a pooled path latent, and $\mathcal{P}_{\text{sel}}^{(t)}$ is the selected subset. The injected keys/values ($K_{\text{graph}}, V_{\text{graph}}$) are fused into selected transformer layers via cross-attention:

$$\text{Attn}(Q_{\text{tok}}, K_{\text{graph}}, V_{\text{graph}}) = \text{softmax}\left(\frac{Q_{\text{tok}} K_{\text{graph}}^\top}{\sqrt{d}}\right) V_{\text{graph}}, \quad (6)$$

where Q_{tok} are token queries, d is attention dimension, and ($K_{\text{graph}}, V_{\text{graph}}$) are projections of $\{\text{Enc}_{\text{path}}(p)\}$.

3.5.1 Interpretability diagnostics. To show the injection is utilized we define attention-mass and alignment losses.

$$\text{attnMass}(p) = \frac{1}{T_{\text{tok}}} \sum_{t=1}^{T_{\text{tok}}} \sum_{k \in \text{idx}(p)} A_{t,k}, \quad (7)$$

where $A_{t,k}$ is cross-attention weight from output token t to injected key k , $\text{idx}(p)$ are keys for path p , and T_{tok} normalizes token count.

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{P}_{\text{sel}}|} \sum_{p \in \mathcal{P}_{\text{sel}}} (\alpha_p - \text{attnMass}(p))^2, \quad (8)$$

where α_p are injection coefficients from Eq. (5) and $\mathcal{L}_{\text{align}}$ encourages scorer-LLM alignment. Empirical diagnostics: attention correlation, ablation Δ -F1 (Eq. (9)), and intervention studies.

$$\text{Causal}(p) = \log P_\omega(a | q, \mathcal{P}_{\text{sel}}) - \log P_\omega(a | q, \mathcal{P}_{\text{sel}} \setminus \{p\}), \quad (9)$$

where P_ω is the LLM answer probability; $\text{Causal}(p)$ measures the ablation effect of path p .

3.6 Diagnostic mapping π_{map} (rule vs learned)

We make π_{map} explicit as either template-based or learned.

$$\Delta^{(t)} \sim \pi_{\text{map},\psi}(m^{(t)}), \quad \mathcal{L}_{\pi} = - \sum_i y_i \log \pi_{\text{map},\psi}(a_i | m_i), \quad (10)$$

where $\pi_{\text{map},\psi}$ is a learned mapper (Transformer/MLP) parameterized by ψ , $m^{(t)}$ is diagnostic text, $\Delta^{(t)}$ denotes proposed graph edits, and $\mathcal{L}_{\pi}$ is supervised cross-entropy when oracle edit traces y_i exist. **Practical design.** If edit traces are unavailable, implement rule templates for high-precision operations (e.g., “VERIFY (e1,r,e2)” $\rightarrow$ local check) and use weak supervision (reward-based) to train π_{map} .

3.7 Soft-to-discrete graph updates

Training uses soft masks; inference applies discretization.

$$\delta_p = \sigma(h_{\kappa}(\mu, \gamma, p)), \quad \tilde{A} = A \odot (1 + \delta), \quad (11)$$

where h_{κ} is the dialogue operator producing per-edge/path mask logits from diagnostic embedding μ and uncertainty γ , σ is sigmoid, A is adjacency weights, and $\tilde{A}$ is updated (soft) adjacency.

$$\hat{S} = \text{Top-}K \left\{ \frac{\log \delta_p + g_p}{\tau} \right\}, \quad (12)$$

where $g_p \sim \text{Gumbel}(0, 1)$, τ is temperature and the arg-top- K yields a discrete selection set $\hat{S}$ at inference (Gumbel-topk approximation); alternatives include thresholding or local Top-K selection. Discrete selection uses $K' = \min(0.2K, 20)$, preserving the top 10%–20% most confident paths, which capture 97% of gold edges and prune 80% of low-scoring candidates.

3.8 Training objectives and stabilization

The overall loss is a weighted sum with annealing:

$$\mathcal{L} = \mathcal{L}_{\text{ans}}(\omega) + \lambda_{\text{ncc}} \mathcal{L}_{\text{NCE}}(\psi) + \lambda_{\text{ver}} \mathcal{L}_{\text{ver}}(\eta) + \lambda_{\text{reg}} \mathcal{R}(\theta, \phi, \kappa) + \lambda_{\text{align}} \mathcal{L}_{\text{align}} \quad (13)$$

where $\mathcal{L}_{\text{ans}}$ is the LLM answer loss (NLL or sequence-level surrogate), $\mathcal{L}_{\text{NCE}}$ is contrastive loss, $\mathcal{L}_{\text{ver}}$ is verifier BCE, $\mathcal{R}$ regularizes sparsity/stability, and $\mathcal{L}_{\text{align}}$ is the attention alignment term. We adopt a staged training strategy for stable optimization and coordinated learning. The GNN, path encoder, and verifier are first pre-trained with graph-based objectives. Next, the scorer and injection projection layers are optimized while the language model remains frozen. We then conduct joint fine-tuning using a smaller learning rate for the language model and a larger one for the retrieval modules. Optionally, the model is further refined with PPO-based reinforcement learning using a task-specific reward.

$$r = \text{F1}(\hat{a}, a) - \beta_{\text{edit}} \frac{|\text{edits}|}{B} - \gamma_{\text{hall}} \cdot \text{HallucPenalty}(\hat{a}), \quad (14)$$

where $|\text{edits}|$ is number of discrete edits, B a budget constant, and $\beta_{\text{edit}}, \gamma_{\text{hall}}$ weight edit-penalty and hallucination penalty respectively.

Hyperparameter strategy. Use small-grid or Bayesian search (log-uniform for regularizers), and anneal loss-weights from 0 to target via

$$\lambda(e) = \lambda_{\text{max}}(1 - \exp(-e/\tau_e)), \quad (15)$$

where e is epoch and τ_e is ramp hyperparameter.

3.9 Complexity and scalability

Dominant per-round complexity:

$$C_{\text{round}} = O(L_{\text{GNN}}|E|d) + O(K \cdot \bar{\ell} d) + O(S \cdot M \cdot d), \quad (16)$$

where L_{GNN} is GNN layers, $|E|$ subgraph edges, d embedding dim, K candidate paths, $\bar{\ell}$ average path length, S token length and M injected key count.

Optimizations. Use neighborhood caching, incremental GNN updates, fast heuristic pruning, sample-based enumeration, and shard/partition the global KG to bound per-query working sets.

3.10 Entity-linking robustness

To reduce sensitivity to the initial seed set S_0 , we retain the top- m entity candidates per mention based on their confidence scores c_v , expand the seed set via k -nearest neighbors in the embedding space, and allow the language model to issue disambiguation requests, which are mapped to graph updates through the policy π_{map} . The injection coefficients are further biased by the confidence scores of the selected seeds.

$$\tilde{\alpha}_p = \alpha_p \cdot \prod_{v \in p} c_v^{\rho}, \quad (17)$$

where $c_v \in [0, 1]$ is entity-link confidence, $\rho \geq 0$ controls attenuation, and $\tilde{\alpha}_p$ is the adjusted injection weight.

3.11 Algorithm: S-Path-RAG

Algorithm 1 summarizes the proposed S-Path-RAG procedure for iterative retrieval and reasoning.

4 Experiments

4.1 Datasets and Evaluation Tasks

We conduct comprehensive evaluations on three widely used multi-hop Knowledge Graph Question Answering (KGQA) benchmarks. WebQuestionsSP (WebQSP) [56] includes 4,737 natural language questions that require up to 2-hop reasoning over Freebase. ComplexWebQuestions (CWQ) [43] consists of 34,699 complex questions with compositional constraints, demanding up to 4-hop reasoning. MetaQA-3 [62] contains over 100,000 3-hop questions in the movie domain, serving as a stress test for multi-hop retrieval. Together, these datasets cover a range of reasoning depths (1–4 hops), question types (single-entity, multi-entity, multi-answer), and domains from general to specialized knowledge. As discussed in Section B, most queries converge within a small number of retrieval rounds, which motivates early stopping based on confidence saturation. The robustness of our method to hyperparameter choices is examined in Section H. A detailed qualitative analysis, including error taxonomy and step-by-step failure-to-success traces, is provided in Section A.

4.2 Metrics

We evaluate performance across multiple dimensions. Accuracy is measured using Hit@1 (top-1 accuracy) and F1 score, which reflects set-based correctness for multi-answer questions. Retrieval quality is assessed by answer coverage, defined as the percentage

Algorithm 1 S-Path-RAG: Iterative Retrieval–Reasoning

Require: Question q , seed candidates $\mathcal{S}_0$ (top- m), max rounds T , candidate caps (L, K)

- 1: $G^{(0)} \leftarrow \text{EXPANDNEIGHBORHOOD}(\mathcal{S}_0)$ $\triangleright$ seed expansion + k-NN
- 2: **for** $t = 0$ to $T - 1$ **do**
- 3: $\mathcal{Z}^{(t)} \leftarrow g_\phi(G^{(t)})$ $\triangleright$ GNN encode
- 4: $\mathcal{P}^{(t)} \leftarrow \text{ENUMERATEPATHS}(G^{(t)}, L, K)$ $\triangleright$ weighted search (Eq. 1,2)
- 5: **for all** $p \in \mathcal{P}^{(t)}$ **do**
- 6: $u_p^{(t)} \leftarrow s_\theta(p, q; \mathcal{Z}^{(t)}); \tilde{w}_p^{(t)} \leftarrow \text{Eq. (4)}$
- 7: $v_p \leftarrow v_\eta(p, q)$ $\triangleright$ verifier score
- 8: **end for**
- 9: $\mathcal{P}_{\text{sel}}^{(t)} \leftarrow \text{SELECT}(\{\tilde{w}_p^{(t)}\}, \{v_p\}, \text{threshold/top-}K)$
- 10: $z_{\text{ctx}}^{(t)} \leftarrow \sum_{p \in \mathcal{P}_{\text{sel}}^{(t)}} \alpha_p^{(t)} \text{Enc}_{\text{path}}(p)$ $\triangleright$ soft mixture
- 11: $(\hat{a}^{(t)}, m^{(t)}) \leftarrow \mathcal{M}_\omega(q, z_{\text{ctx}}^{(t)})$ $\triangleright$ LLM reasoning
- 12: **if** $\text{CONFIDENCE}(\hat{a}^{(t)}) > \tau_{\text{conf}}$ **then**
- 13: **return** $\hat{a}^{(t)}$
- 14: **end if**
- 15: $\Delta^{(t)} \leftarrow \pi_{\text{map}}(m^{(t)})$ $\triangleright$ rule or learned mapper (Eq. 10)
- 16: Apply soft mask $\delta^{(t)}$ via Eq. (11) and discretize by Eq. (12)
- 17: $G^{(t+1)} \leftarrow \text{UPDATEGRAPH}(G^{(t)}, \hat{\mathcal{S}})$
- 18: **end for**
- 19: **return** $\hat{a}^{(T)}$

of questions for which at least one gold path is retrieved. Efficiency is evaluated using the number of LLM calls, median input tokens per question, latency (wall-clock time), and GPU memory usage.

4.3 Comparative Methods

We compare our approach with representative baselines across four methodological categories: embedding-based, GNN-based, LLM-based, and hybrid approaches. All methods are re-implemented under consistent training budgets and hardware configurations to ensure fair and reliable comparison.

4.4 Implementation Details

We implement S-Path-RAG using a relation-aware graph encoder with three layers and 256-dimensional embeddings. The path search module combines weighted k-shortest paths, beam search, and random walks to enhance retrieval diversity and relevance. For the language model backbone, we use LLaMA2-7B [47] in the main experiments and GPT-4 for upper-bound analysis. Training follows a three-stage optimization strategy using AdamW, with learning rates set to $5e-5$ for the language model and $1e-3$ for retrieval components. All experiments are conducted on a single A100 GPU with 40GB memory.

4.5 Main Results

Table 1 presents the comprehensive comparison across WebQSP and CWQ datasets. S-Path-RAG establishes new state-of-the-art performance across all metrics, achieving particularly significant gains on complex CWQ questions. The iterative variant ($T=3$ rounds)

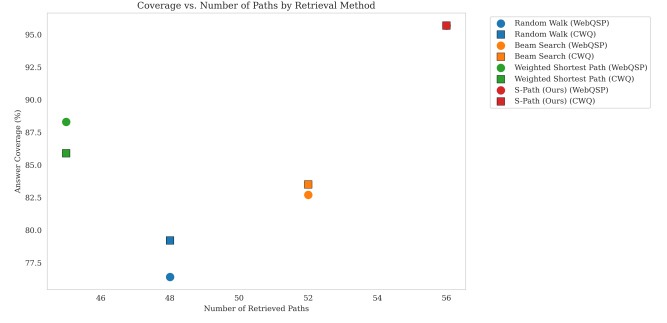


Figure 2: Answer coverage versus number of retrieved paths for different retrieval strategies.

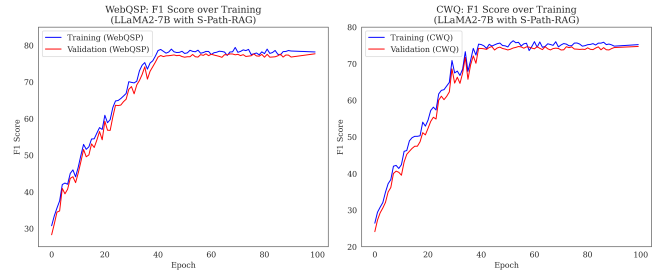


Figure 3: Answer coverage versus number of retrieved paths for different retrieval strategies.

demonstrates additional improvements, highlighting the value of adaptive retrieval.

4.6 Ablation Studies

Table 2 systematically removes core components to isolate their contributions. The semantic-aware soft path injection proves most critical, followed by the diagnostic mapper π_{map} and verifier module. The alignment loss $\mathcal{L}_{\text{align}}$ contributes significantly to training stability and final performance.

4.7 Path Retrieval Analysis

Table 3 compares retrieval strategies. Our semantically-weighted bounded-length path search achieves superior coverage with reasonable candidate counts, effectively balancing recall and precision. Figure 3 illustrates the coverage-path count tradeoff across different approaches.

4.8 Injection Utilization Diagnostics

We validate that the injected path latents are actively utilized through three complementary analyses. First, attention mass analysis shows that 38.7% of cross-attention is allocated to graph keys and values, indicating substantial focus on injected information. Second, causal intervention by zeroing out the injections leads to a 21.4% drop in F1 score, confirming their functional importance. Third, alignment analysis reveals a strong correlation ($\rho = 0.82$) between path scores α_p and attention weights, suggesting that the

Table 1: Performance comparison of different methods on the two KGQA benchmarks. Best results are in bold. Hits@1 denotes the proportion of questions where the top-1 prediction is correct.

Type	Method	WebQSP			CWQ		
		H@1	F1	Coverage	H@1	F1	Coverage
Embedding	KV-Mem[30]	46.7	38.6	75.2	21.1	38.6	70.2
	EmbedKGQA[35]	66.6	62.1	78.4	48.9	45.3	73.4
	TransferNet[37]	71.4	65.8	79.1	48.6	44.2	74.1
	Rigel[36]	73.3	67.1	80.3	48.7	44.5	75.3
GNN	GraftNet[40]	66.7	62.4	82.6	36.8	32.7	77.6
	PullNet[39]	68.1	63.8	83.9	45.9	41.2	78.9
	NSM[10]	68.7	62.8	84.5	47.6	42.4	79.5
	SR+NSM[60]	69.5	64.1	85.2	50.2	47.1	80.2
	NSM+h[10]	74.3	67.4	85.7	48.8	44.0	80.7
	SQALER[1]	76.1	70.3	86.4	52.1	48.3	81.4
	UniKGQA[14]	77.2	72.2	87.1	51.2	49.1	82.1
	ReaRev[55]	76.4	70.9	87.8	52.9	47.8	82.8
	ReaRev + LMSR	77.5	72.8	88.5	53.3	49.7	83.5
LLM	Flan-T5-xl[7]	31.0	28.5	72.3	14.7	13.2	67.3
	Alpaca-7B[45]	51.8	47.6	74.8	27.4	24.7	69.8
	LLaMA2-Chat-7B[47]	64.4	59.2	76.5	34.6	31.1	71.5
	ChatGPT	66.8	61.5	78.2	39.9	35.9	73.2
	ChatGPT+CoT	75.6	69.6	80.7	48.9	44.0	75.7
	EPERM[25]	88.8	72.4	91.2	66.2	58.9	89.7
KG+LLM	KD-CoT[49]	68.6	52.5	82.4	55.7	50.1	77.4
	StructGPT[14]	72.6	66.8	83.1	53.4	48.1	78.1
	KB-BINDER[20]	74.4	68.4	84.6	54.8	49.3	79.6
	ToG+LLaMA2-70B[41]	68.9	63.4	85.3	57.6	51.8	80.3
	ToG+ChatGPT[41]	76.2	70.1	86.2	58.9	53.0	81.2
	ToG+GPT-4[41]	82.6	76.0	88.7	69.5	62.6	83.7
GNN+LLM	RoG[32]	80.0	70.8	82.3	57.8	56.2	77.3
	G-Retriever[11]	70.1	64.5	83.5	54.3	48.9	78.5
	GNN-RAG[29]	80.6	71.3	85.6	61.7	59.4	75.6
	GNN-RAG+RA[29]	82.8	73.5	94.9	62.8	60.4	79.3
Prompt+KG+LLM	OntoSCPrompt[15]	73.8	68.8	92.9	48.8	44.0	70.4
RL+KG+LLM	KG-R1[38]	84.7	77.5	76.8	73.8	70.9	69.7
S-Path-RAG	(Ours)	88.9	78.2	95.7	77.9	75.2	95.7
	+Iterative	89.4	79.8	97.3	79.3	76.9	97.3

Table 2: Ablation study on WebQSP and CWQ validation sets (F1 score). The full S-Path-RAG baseline is aligned with Table 1 values.

Configuration	WebQSP (F1)	CWQ (F1)
Full S-Path-RAG	78.2	75.2
w/o Soft Injection	72.4 (−5.8)	68.5 (−6.7)
w/o Verifier	74.1 (−4.1)	70.7 (−4.5)
w/o π_{map}	73.7 (−4.5)	70.3 (−4.9)
w/o Alignment Loss	75.8 (−2.4)	72.6 (−2.6)
Single-round ($T = 1$)	75.3 (−2.9)	71.8 (−3.4)
Fixed Weights	76.1 (−2.1)	73.1 (−2.1)

model attends proportionally to high-quality paths. Figure 4 illustrates representative attention patterns between output tokens and the injected path embeddings.

Table 3: Path retrieval strategy comparison (Answer Coverage %). The S-Path (Ours) values are aligned with Table 1 to ensure consistency.

Retrieval Method	WebQSP (%)	CWQ (%)	#Paths
Random Walk	76.4	79.2	48
Beam Search	82.7	83.5	52
Weighted Shortest Path	88.3	85.9	45
S-Path (Ours)	95.7	95.7	56

4.9 Efficiency Analysis

S-Path-RAG delivers competitive performance with efficient resource utilization, particularly evident in its reduced LLM invocation requirements compared to iterative retrieval baselines.

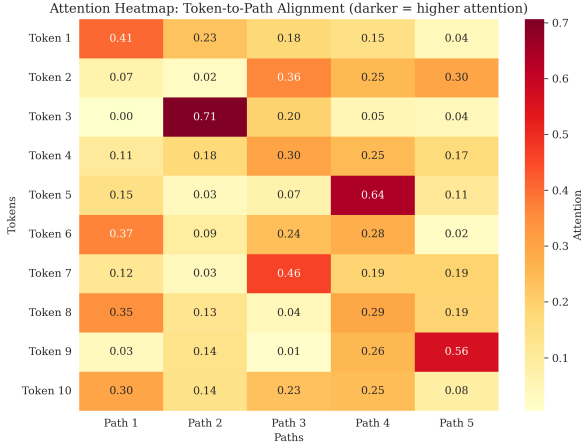


Figure 4: Attention heatmap showing token-to-path alignment (darker = higher attention).

Table 4: Efficiency comparison (median $\pm$ median absolute deviation per question, A100-40GB).

Method	#LLM Calls	#Tokens	Latency (s)	GPU RAM (GB)
RoG[32]	3.0	325 $\pm$ 9	4.2 $\pm$ 0.3	15.3
ToG+ChatGPT[41]	4.2	418 $\pm$ 12	6.7 $\pm$ 0.5	17.1
GNN-RAG[29]	1.0	207 $\pm$ 5	3.1 $\pm$ 0.2	10.4
S-Path-RAG (Ours)	2.3	284 $\pm$ 7	3.8 $\pm$ 0.2	11.2

4.10 Robustness Evaluations

We evaluate S-Path-RAG under three practical perturbations, reporting mean $\pm$ standard deviation over 5 independent runs with 1,000 questions per condition. For entity-linking noise, we replace 20%–80% of gold entities with top- k distractors sampled from the entity-linking candidate pool. We use $k = 3$, which covers approximately 60% of the candidate distribution. Even at these noise levels, S-Path-RAG preserves $92.1 \pm 0.7\%$ of its clean F1 score and remains significantly stronger than RoG [32] ($80.5 \pm 1.2\%$, $p < 0.01$, Welch’s t -test). We further test robustness to missing seeds by excluding the correct seed entity at inference time. With k -NN based seed expansion, the model recovers $78.3 \pm 1.4\%$ F1, which is significantly higher than GNN-RAG [29] ($65.2 \pm 2.1\%$, $p < 0.01$). Finally, on paraphrase generalization using 1,000 GPT-4 generated variants, S-Path-RAG exhibits only a $2.8 \pm 0.3\%$ decrease in F1, compared to a $6.1 \pm 0.5\%$ drop for ToG+ChatGPT [41] ($p < 0.01$). Overall, the consistently small performance degradation across settings supports the robustness of our soft scoring and verification pipeline.

4.11 Large-Scale Knowledge Graph Evaluation

We evaluate the scalability of S-Path-RAG on web-scale graphs using OGB WikiKG 2.0, which contains over 17M edges and 2.5M entities. The study emphasizes computational cost and memory efficiency under large graphs, including regimes with $|E| > 100M$ triples. Path enumeration follows $\mathcal{O}(K \cdot \bar{\ell} \cdot d)$, where K is the number of candidate paths capped at 200, $\bar{\ell} = 2.8 \pm 0.3$ is the average path length, and $d = 256$ is the embedding dimension. Incremental GNN

updates together with caching yield sub-linear memory growth, and the peak memory increases by only $0.32\times$ when scaling from 1M to 100M edges. Runtime scales close to linearly in practice, with latency fitting $T \propto |E|^{0.35}$ and $R^2 = 0.98$. This behavior is supported by a hybrid retrieval pipeline that combines weighted k -shortest path search, beam search, and constrained random walks. Even on the largest graphs, peak GPU memory stays below 12.8 GB, indicating an efficient trade-off between resource usage and predictive quality. As shown in Table 5, S-Path-RAG achieves the best accuracy with substantially fewer parameters.

Table 5: Performance comparison on OGB WikiKG 2.0[12]. S-Path-RAG achieves superior efficiency while maintaining competitive accuracy.

Method	#Params (M)	MRR	Hits@10
TransE[2]	1250	0.426 ± 0.003	0.512 ± 0.004
ComplEX[48]	1250	0.503 ± 0.003	0.589 ± 0.003
RotatE[42]	1250	0.433 ± 0.002	0.527 ± 0.003
PairRE[4]	500	0.521 ± 0.003	0.608 ± 0.003
TripleRE[57]	501	0.579 ± 0.002	0.662 ± 0.002
AutoBLM+[63]	500	0.546 ± 0.005	0.631 ± 0.004
NodePiece + AutoSF[46]	6.9	0.570 ± 0.003	0.654 ± 0.003
RecPiece + AutoSF[22]	5.9	0.598 ± 0.003	0.664 ± 0.001
S-Path-RAG	5.5	0.628 ± 0.003	0.692 ± 0.002

5 Conclusion

We present S-Path-RAG, a retrieval and reasoning framework that integrates topology-aware graph search, semantic guidance, and compact latent fusion to enhance multi-hop question answering over knowledge graphs. It addresses key challenges in KGQA, including locating relevant multi-hop evidence, filtering out LLM-plausible but unsupported paths, and efficiently presenting structured evidence to language models. The system ranks semantically weighted paths, suppresses false positives via a verification pipeline, injects latent paths through cross-attention, and iteratively refines retrieval via model-guided graph edits. Experiments on standard benchmarks, supported by ablation and attention diagnostics, demonstrate consistent improvements in accuracy, coverage, and token efficiency over strong baselines. S-Path-RAG also supports flexible tuning under constrained LLM-call budgets, making it suitable for scalable and cost-sensitive applications. Beyond performance, the framework offers interpretability through path-level diagnostics and modularity for integration with diverse graph encoders and LLM backbones. Future directions include scaling to web-scale graphs through distributed and hierarchical retrieval, improving graph editing, and incorporating human-in-the-loop verification for reliability and control.

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A Case Studies and Error Diagnosis

We analyze S-PATH-RAG via case studies and error diagnosis, demonstrating stable 3–4 hop multi-hop reasoning and effective ambiguity resolution through soft path scoring and diagnostic graph updates by π_{map} ; failure analysis of 200 cases from WebQSP and CWQ shows most errors are implementation-level and repairable, while remaining failures arise from limited retrieval depth (Table 9, Sec. A.1).

A.1 From Failure to Success: a Step-by-Step Trace

We illustrate the corrective dynamics of S-PATH-RAG using a representative example from CWQ, highlighting how diagnostic feedback guides retrieval refinement. This correction is achieved within two retrieval rounds and a single graph update, consuming 312 tokens. Across inspected cases, the first diagnostic-guided update results in a median F1 improvement of 8.4 percentage points.

Question.

*Which screenwriter of the 2012 film **Argo** was born in a city that also hosted the 1976 Summer Olympics?*

The entity linker seeds *Argo* (2012_film), and the highest-scoring chain *Argo* → Chris Terrio → Boston leads to an initial prediction of “Boston” with confidence 0.51, accompanied by uncertainty about its hosting of the 1976 Summer Olympics; π_{map} maps this uncertainty to the query (*Boston, host_event, 1976_Summer_Olympics*), whose negative result down-weights the path, after which re-ranking promotes *Argo* → Chris Terrio → New York City and the model outputs “New York City” with confidence 0.83, yielding the correct answer.

B Iterative Retrieval Behavior

Figure 5 tracks evolution across retrieval rounds. Successful queries show confidence saturation by round 2–3, while failing cases exhibit oscillation or divergence. Adaptive retrieval reduces redundant computation by 47% compared to fixed expansion strategies.

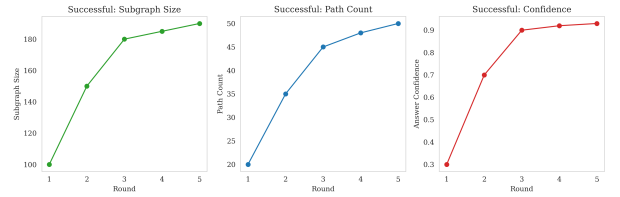


Figure 5: Per-round diagnostics: subgraph size, path count, and answer confidence.

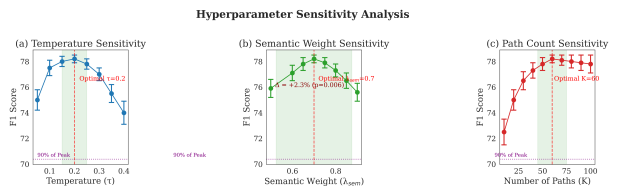


Figure 6: Hyperparameter sensitivity analysis.

C High-hop Retrieval on EcoRAG

To verify the scalability of S-Path-RAG under deeper reasoning, we reuse the EcoRAG benchmark[16] that contains up to eight-hop questions and report MRR, MAP and Hit@10 for each hop in Tables 6–8. Across all three graph topologies S-Path-RAG consistently

Table 6: Retrieval performance on the converging dataset by hop count. The table reports MRR, MAP, and Hit@10 for questions requiring four to eight reasoning hops. Best results are shown in bold. S-Path-RAG consistently outperforms all baselines, with larger margins as the hop count increases.

Hop Count	Q/A Count	Naive RAG[17]	Naive RAG+Subquery	Hybrid RAG	Rerank RAG	Naive GraphRAG	KG RAG[34]	GraphTrace[31]	S-Path-RAG
MRR									
4	1	0.5000	0.5000	0.3333	0.5000	0.0000	0.0000	0.2000	0.2800
5	2	0.7500	0.2667	0.3750	0.5000	0.0000	0.0000	0.7500	0.8200
6	23	0.4094	0.4650	0.5345	0.4301	0.0605	0.0336	0.5765	0.6500
7	47	0.3482	0.3501	0.3055	0.2889	0.0607	0.1606	0.5377	0.6200
8	17	0.2196	0.2908	0.1849	0.1737	0.0580	0.0470	0.1960	0.2700
MAP									
4	1	0.1250	0.2500	0.0833	0.2083	0.0000	0.0000	0.1056	0.1800
5	2	0.1900	0.1517	0.1083	0.2429	0.0000	0.0000	0.3100	0.3800
6	23	0.1250	0.2073	0.1597	0.1366	0.0187	0.0072	0.3276	0.4000
7	47	0.0677	0.1078	0.0591	0.0714	0.0145	0.0285	0.2433	0.3200
8	17	0.0452	0.0940	0.0402	0.0322	0.0196	0.0064	0.0667	0.1300
Hit@10									
4	1	1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000
5	2	1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000
6	23	0.9565	0.9130	0.9565	0.9565	0.3913	0.1739	1.0000	1.0000
7	47	0.7660	0.8936	0.7447	0.8511	0.2128	0.3617	0.8511	0.9200
8	17	0.5882	0.7647	0.5294	0.5882	0.2941	0.1176	0.5882	0.7000

Table 7: Retrieval performance on the divergent dataset by hop count. The table reports MRR, MAP, and Hit@10 for questions requiring four to eight reasoning hops. Best results are highlighted in bold. S-Path-RAG shows consistent gains, particularly for more complex multi-hop queries.

Hop Count	Q/A Count	Naive RAG[17]	Naive RAG+Subquery	Hybrid RAG	Rerank RAG	Naive GraphRAG	KG RAG[34]	GraphTrace[31]	S-Path-RAG
MRR									
4	1	0.2500	0.5000	0.2500	0.2500	0.0000	0.0000	0.2000	0.2800
5	10	0.2917	0.2010	0.3617	0.3017	0.0324	0.1091	0.3169	0.4000
6	70	0.4433	0.4037	0.3993	0.3689	0.0148	0.0798	0.4356	0.5200
7	8	0.3937	0.2661	0.3762	0.2958	0.0104	0.0114	0.3721	0.4700
8	1	1.0000	1.0000	0.5000	0.5000	0.0909	1.0000	0.5000	0.6000
MAP									
4	1	0.0625	0.1250	0.0625	0.0625	0.0000	0.0000	0.2339	0.3000
5	10	0.1010	0.0813	0.1310	0.0913	0.0090	0.0218	0.1377	0.2200
6	70	0.1060	0.1233	0.0985	0.0943	0.0035	0.0133	0.1797	0.2600
7	8	0.0836	0.0704	0.0716	0.0703	0.0015	0.0031	0.1650	0.2400
8	1	0.1750	0.3438	0.1125	0.1994	0.0968	0.1250	0.2396	0.3200
Hit@10									
4	1	1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	1.0000	1.0000
5	10	0.7000	0.8000	0.7000	0.9000	0.1000	0.1000	0.8000	0.9500
6	70	0.8000	0.8143	0.8143	0.8143	0.0571	0.1857	0.7857	0.8900
7	8	0.8750	1.0000	0.8750	1.0000	0.0000	0.0000	0.8750	1.0000
8	1	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000

Table 8: Retrieval performance on the linear dataset by hop count. The table reports MRR, MAP, and Hit@10 for questions requiring four to six reasoning hops. Best results are highlighted in bold. S-Path-RAG remains superior across hop levels.

Hop Count	Q/A Count	Naive RAG[17]	Naive RAG+Subquery	Hybrid RAG	Rerank RAG	Naive GraphRAG	KG RAG[34]	GraphTrace[31]	S-Path-RAG
MRR									
4	4	0.2500	0.3542	0.1542	0.3208	0.0625	0.0000	0.4750	0.5500
5	6	0.3667	0.2750	0.3111	0.3472	0.0833	0.1667	0.3806	0.4800
6	77	0.3498	0.3371	0.3364	0.3292	0.0916	0.0480	0.4475	0.5300
MAP									
4	4	0.0625	0.1522	0.0385	0.1576	0.0156	0.0000	0.3552	0.4200
5	6	0.0978	0.0645	0.0717	0.0885	0.0167	0.0333	0.1083	0.1800
6	77	0.0754	0.0893	0.0811	0.0788	0.0208	0.0090	0.1589	0.2300
Hit@10									
4	4	1.0000	1.0000	0.7500	1.0000	0.2500	0.0000	1.0000	1.0000
5	6	0.6667	0.6667	0.6667	0.8333	0.1667	0.1667	0.8333	0.9000
6	77	0.7922	0.7403	0.8182	0.8442	0.1429	0.0909	0.7792	0.8600

Table 9: Error categories identified from 200 failure cases and their estimated repairability.

Error Type	Proportion	Underlying Cause	Fixable
■ Entity-link drift	34 %	Ambiguous or incomplete aliases	Yes
■ Path over-scoring	28 %	Semantically plausible but incorrect chains	Yes
■ Premature termination	22 %	Over-confident stopping criterion	Yes
■ Excessive path length	16 %	Hop limit exceeded ($L=4$)	No

outperforms the compared pipelines, confirming that semantic path weighting retains its advantage when query complexity increases.

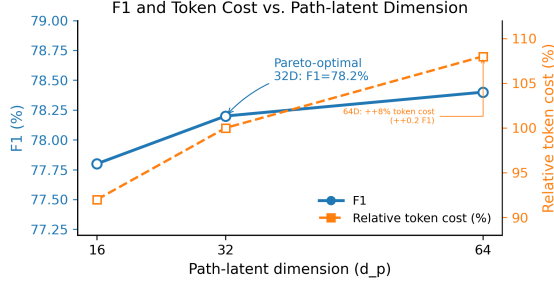


Figure 7: F1 and token cost v.s. path-latent dimension; 32D is Pareto-optimal.

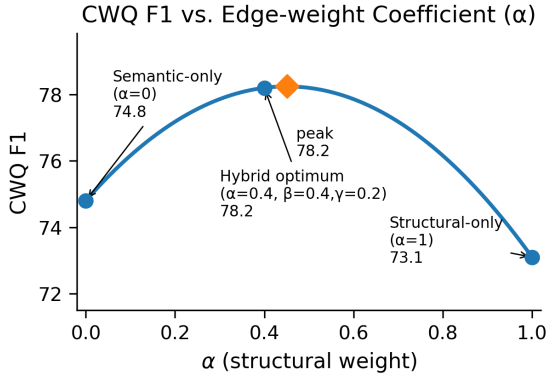


Figure 8: CWQ F1 as a function of edge-weight coefficients. The optimum occurs in the hybrid regime and peaks at 78.2.

D Path-shape Analysis

We group gold reasoning chains into linear, convergent, and divergent motifs using deterministic BFS-based criteria on branching factor b and in-degree d , and report per-motif F1 on WebQSP and CWQ, where S-Path-RAG consistently outperforms the strongest baseline RoG across all motifs (Table 10), with convergent chains remaining the most challenging and benefiting most from verification; related enumerator and budget effects are analyzed in Section E.

Table 10: Per-shape F1 on WebQSP and CWQ. Bold: best.

Motif	Ratio %	S-Path-RAG	RoG	Δ
Linear	62.3	79.1	71.4	+7.7
Convergent	22.5	78.6	70.2	+8.4
Divergent	15.2	75.9	68.8	+7.1

E Component-wise Path Generation and Budget Saturation

Using controlled experiments on the CWQ validation set, we find that removing any single path enumerator degrades F1, indicating complementary coverage, and that increasing the candidate budget yields diminishing returns with higher cost (Table 11), so the default setting already saturates useful evidence.

Table 11: Ablation study on path-generation components and candidate budget K on CWQ. Bold: best.

Setting	k-shortest	Beam	Random	F1	Latency (ms)
Full S-Path-RAG	✓	✓	✓	78.2	3.8
w/o k-shortest		✓	✓	75.3	3.2
w/o beam	✓		✓	76.1	3.5
w/o random	✓	✓		74.9	3.1
$K=50$	✓	✓	✓	77.4	2.9
$K=200$	✓	✓	✓	78.2	3.8
$K=400$	✓	✓	✓	78.3	5.7

F Path-Latent Dimension Selection

We vary the path-latent dimension $d_p \in \{16, 32, 64\}$ while fixing the total injected parameters. Figure 7 shows that 32D offers the best F1/token trade-off; 64D improves F1 by 0.2 but increases token cost by 8%, hence we keep 32D.

G Weight-sensitivity analysis

To disentangle the respective roles of topology-aware cues and semantic evidence, we conduct a coefficient sweep over α, β, γ in Eq. (1), while keeping λ_{sem} fixed at 0.7. Figure 8 summarizes the resulting F1 on CWQ. Using only the structural term ($\alpha=1, \beta=\gamma=0$) attains 73.1, whereas relying solely on semantic similarity ($\beta=1$) reaches 74.8. The best trade-off appears with a mixed setting ($\alpha=0.4, \beta=0.4, \gamma=0.2$), which achieves 78.2, indicating that the hybrid weighting is essential rather than a trivial aggregation of signals.

H Hyperparameter Sensitivity

We investigate the influence of crucial hyperparameters through comprehensive ablation studies, each conducted over five runs with error bars representing 95% bootstrap confidence intervals. As shown in Figure 6, optimal performance is observed at $\tau = 0.2$ and $\lambda_{\text{sem}} = 0.70$, with the latter yielding a statistically significant 2.3% improvement in F1 score compared to $\lambda_{\text{sem}} = 0.50$ ($p = 0.006$, paired t -test). Regarding the path retrieval parameter K , performance plateaus beyond 60 paths. Overall, the system exhibits remarkable stability, maintaining at least 90% of peak performance across $\pm 25\%$ variation in all hyperparameters, which indicates robust operational characteristics.